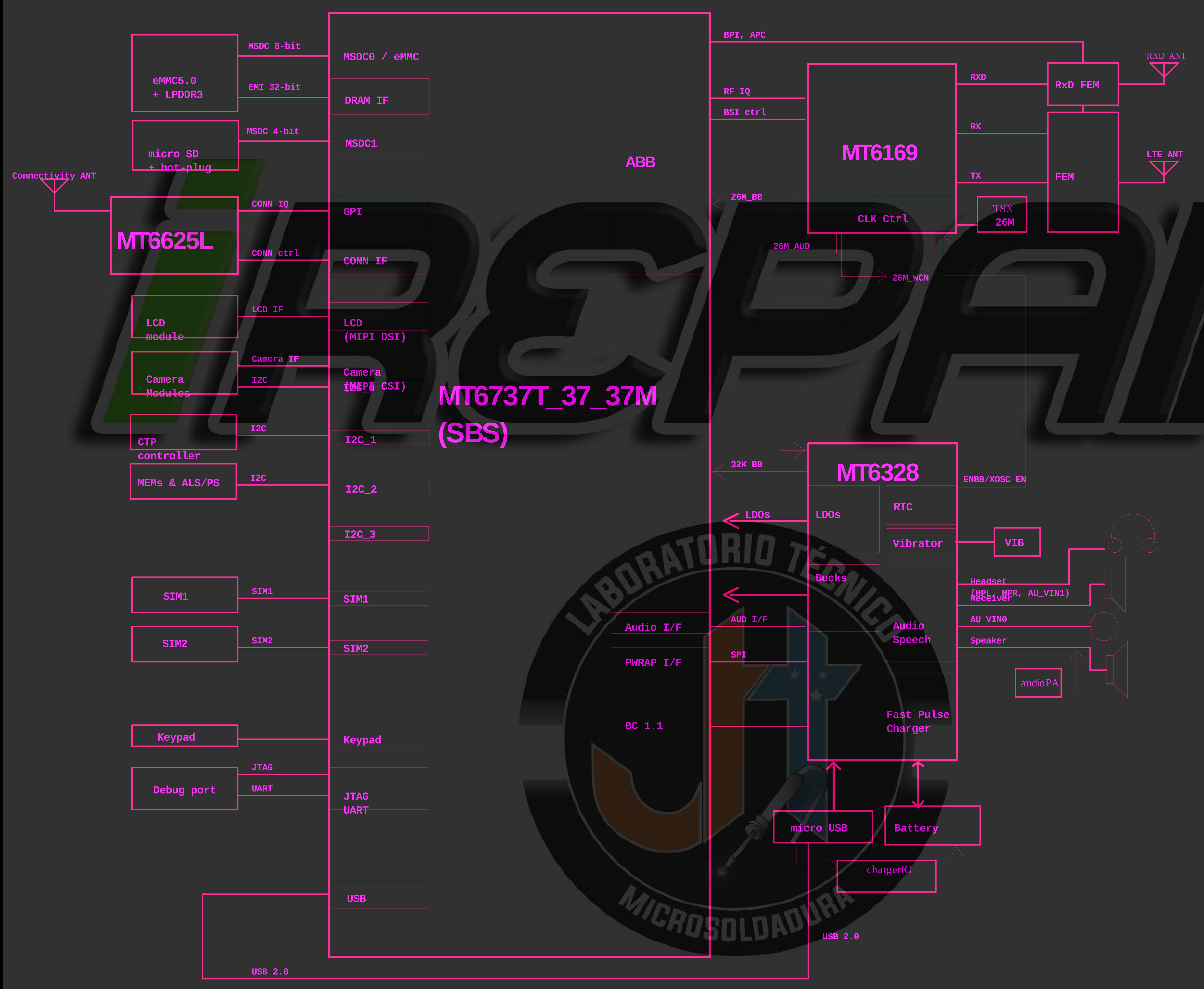
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Motorola

XT1758 Block_diagram&SCH

Project : MT6737T_37_37M REF_SCH TOP LEVEL



DRAWN: <Drawn By>	DATED: <Drawn Date>	COMPANY: Motorola			
CHECKED: <Checked By>	DATED: <Checked Date>	TITLE: XT1758			
QUALITY CONTROL: <QC By>	DATED: <QC Date>	CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>		SHEET: 1 OF 21	

I2C	Function	I2C Spec.	Budget Timing	I2C Slave Address (7-bit mode)
I2C-0	Rear Camera - 8M	400 Kbps	Yes.	Rear camera (IMX135) I2C address: 0X10 (Write:0x20, Read:0x21) AF driver (DW9714A) I2C address: 0x0C (Write:0x18 , Read:0x19)
	Front Camera-2M	400 Kbps	Yes.	Front camera (GC2355) I2C address: 0x3C (Write:0x78, Read:0x79)
	charger IC			charger IC (FAN54015) I2C address: 0x6A (Write:0xD4, Read:0xD5)
I2C-1	CTP	400 Kbps	Yes.	GT1151 / CTP I2C address: 0X5D (Write:0xBA, Read:0xBB) or 0x14 (Write:0x28, Read:0x29)
I2C-2	LCD_DC-DC			DC-DC (KTD2151EUO-TR) I2C Address: 0x3E (Write:0x7C, Read:0x7D)
	G-Sensor	400 Kbps	Yes.	G-Sensor (BMA253) I2C Address: 0x18 (Write:0x30, Read:0x31)
I2C-3				

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
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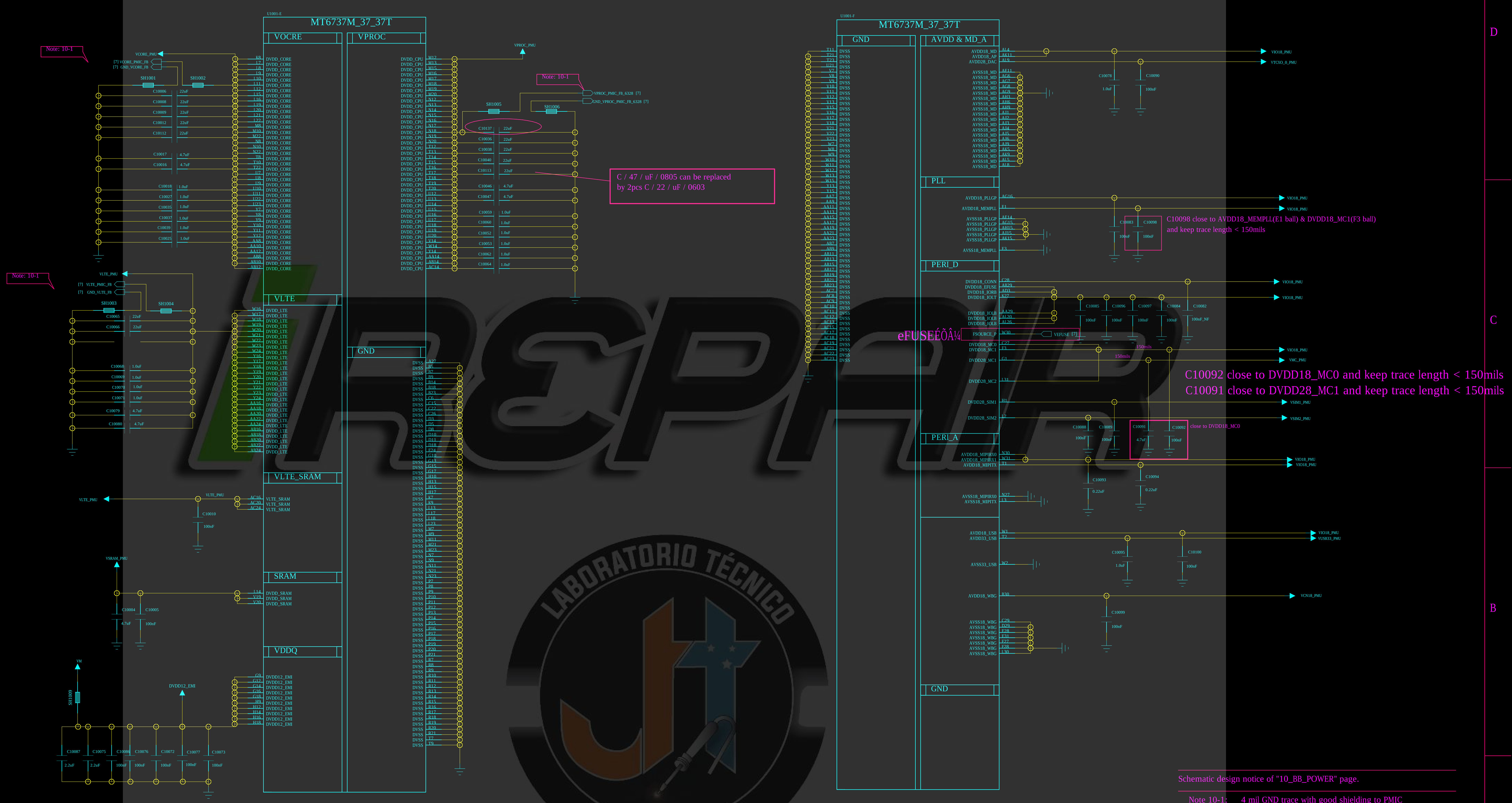
COMPANY: Motorola

TITLE: XT1758

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A1	<Drawing Number>	1_13

SCALE: <Scale> SHEET: 2 of 21

MT6737M



Schematic design notice of "10_BB_POWER" page.

Note 10-1: 4 mil GND trace with good shielding to PMIC (Differential)

Note 10-3: FSOURCE_P(EFUSE)
(1)FSOURCE_P EFUSE power(VEFUSE) should be only for EFUSE usage(not share with other application)
(2)W/I EFUSE program, VEFUSE need 1uF bypass cap (pls refer to ¡§LDO output voltage/current table¡¨)
(3)W/O EFUSE program, VEFUSE bypass cap should be NC.

COMPANY:	Motorola
TITLE:	XT1758
CODE: <Code>	SIZE: A1
DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>	SHEET: 3 of 21
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CHECKED: <Checked By>	DATED: <Checked Date>
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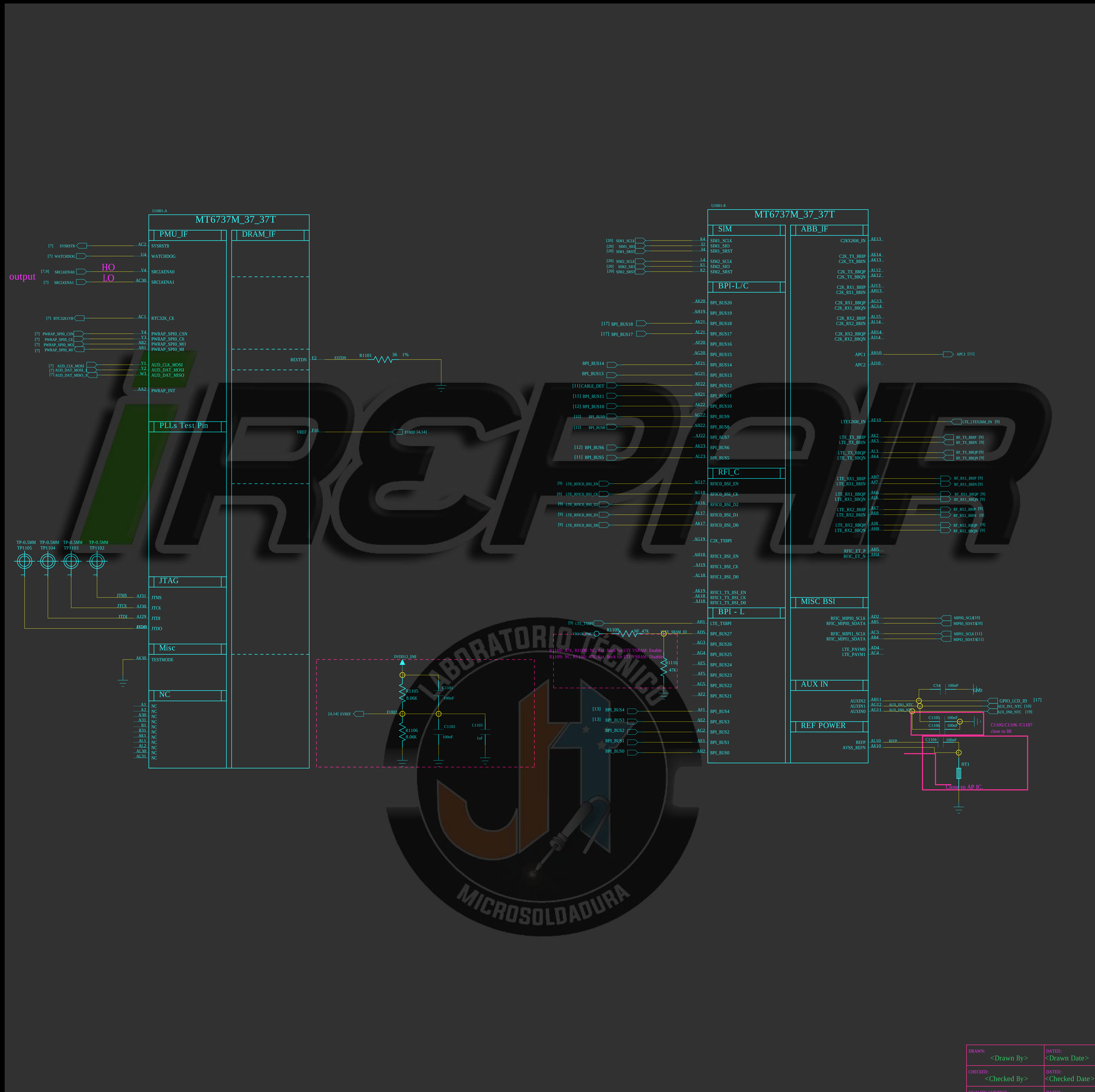
MT6737M



REVISION RECORD

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COMPANY:	Motorola		
TITLE:	XT1758		
CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>		SHEET: 4	21

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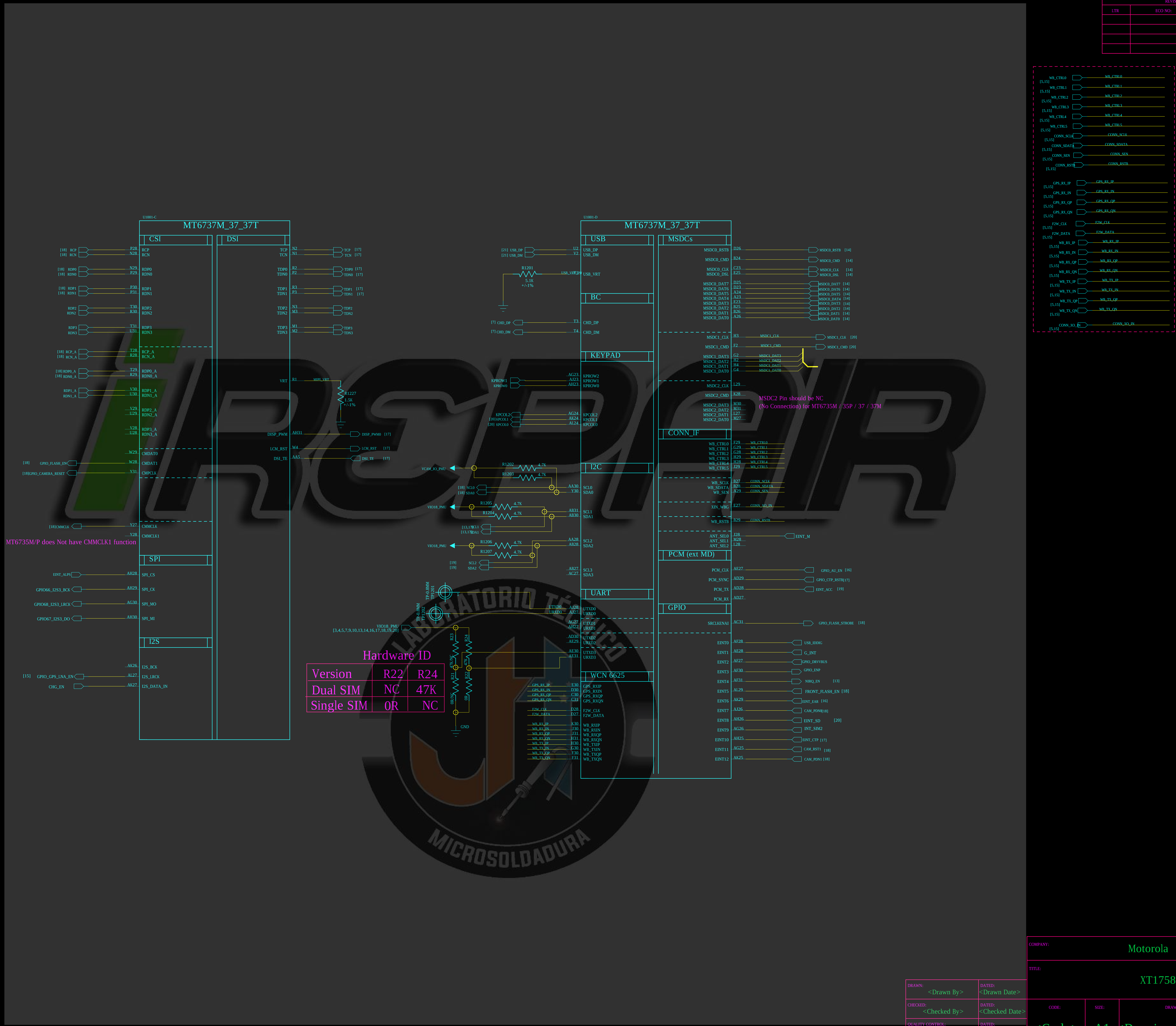
MT6737M



Hardware ID

Version	R22	R24
Dual SIM	NC	47K
Single SIM	0R	NC

COMPANY:	Motorola
TITLE:	XT1758
CODE:	<Code>
SIZE:	A1
DRAWING NO:	<Drawing Number>
REV:	1_13
SCALE:	<Scale>
SHEET:	5 of 21
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DATED:	<Drawn Date>
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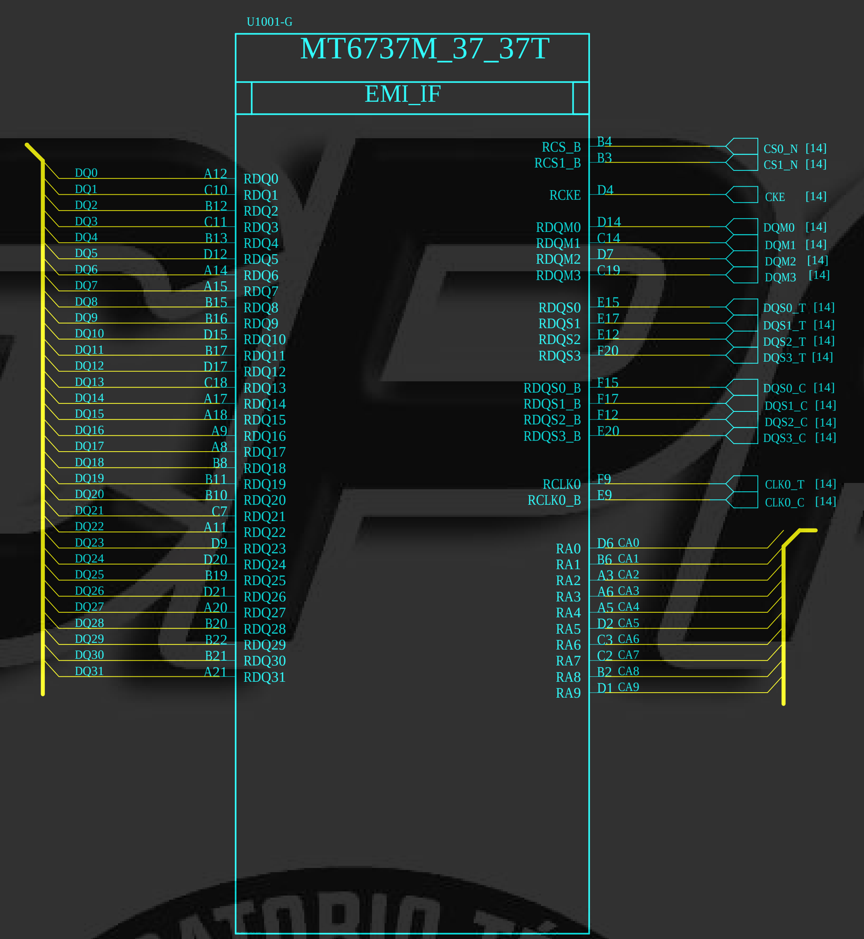
DATED:	<Release Date>

MT6737M

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



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QUALITY CONTROL: <QC By>	DATED: <QC Date>		
RELEASED: <Released By>	DATED: <Release Date>		

COMPANY: Motorola			
TITLE: XT1758			
CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>		SHEET: 6 of 21	

input
output
Please refer to latest MMD design
Close to MT6328
C2040 close to AVDD22_LDO1/2/3/4 balls
UVLO_VTH R RANGE:
200KOHM ~ 250KOHM
Audio
single via to GND
plane directly
C2027/ C2038/ C2058 flying cap & holding cap :
4.7uF for 16ohm reciver
2.2uF for 32ohm reciver
To improve noise level, connect to audio jack first,
and then connect to GND
Close to PMIC
Charger
BC1.1/1.2
Gauge
GND shielding
REVISION RECORD
Motorola
XT1758
<Drawn By>
<Drawn Date>
<Checked By>
<Checked Date>
<QC By>
<QC Date>
<Released By>
<Release Date>
<Code>
A1
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1_13
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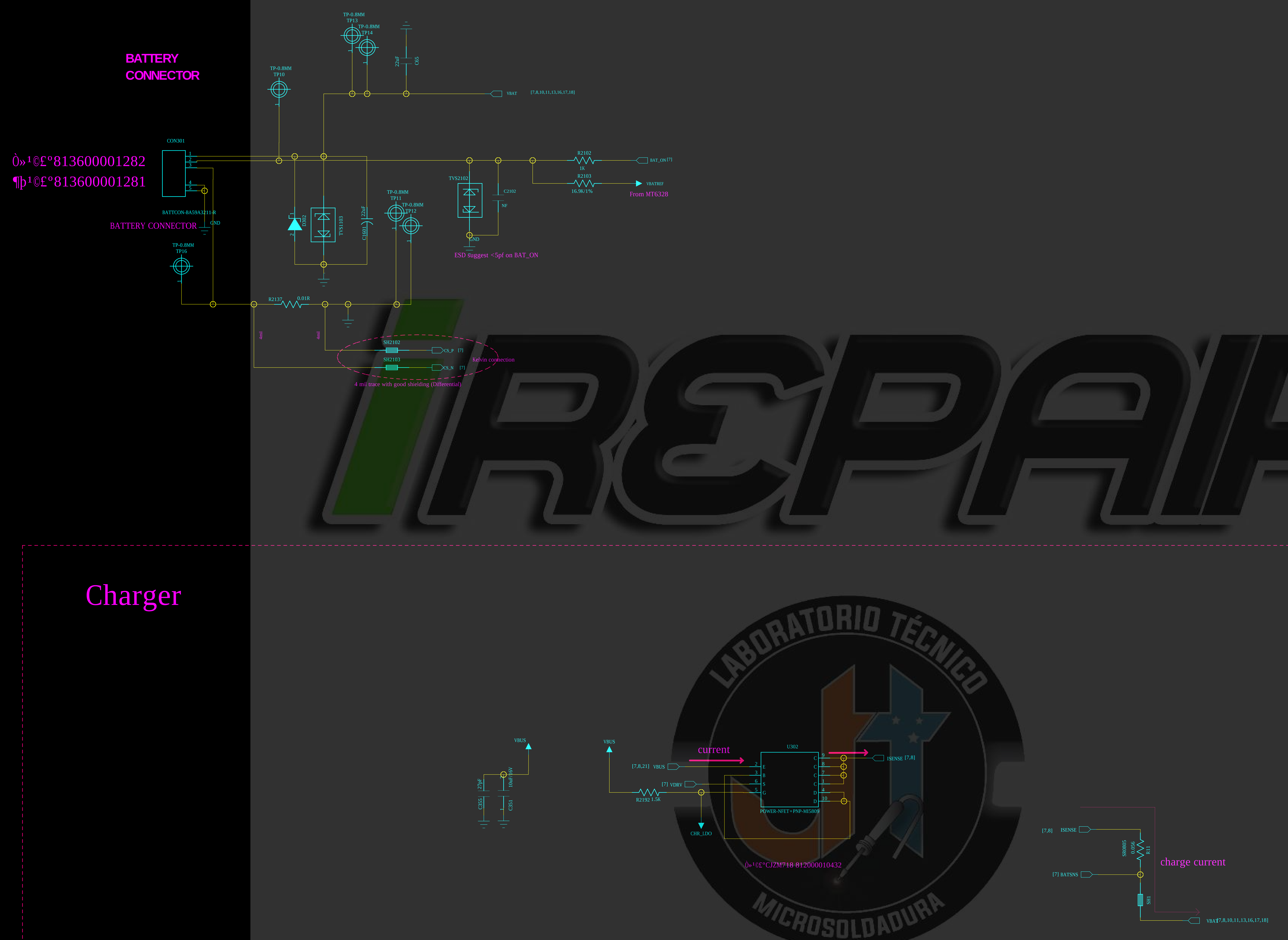
21



REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

COMPANY: Motorola

TITLE: XT1758

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CHECKED: <Checked By>	DATED: <Checked Date>
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RELEASED: <Released By>	DATED: <Release Date>

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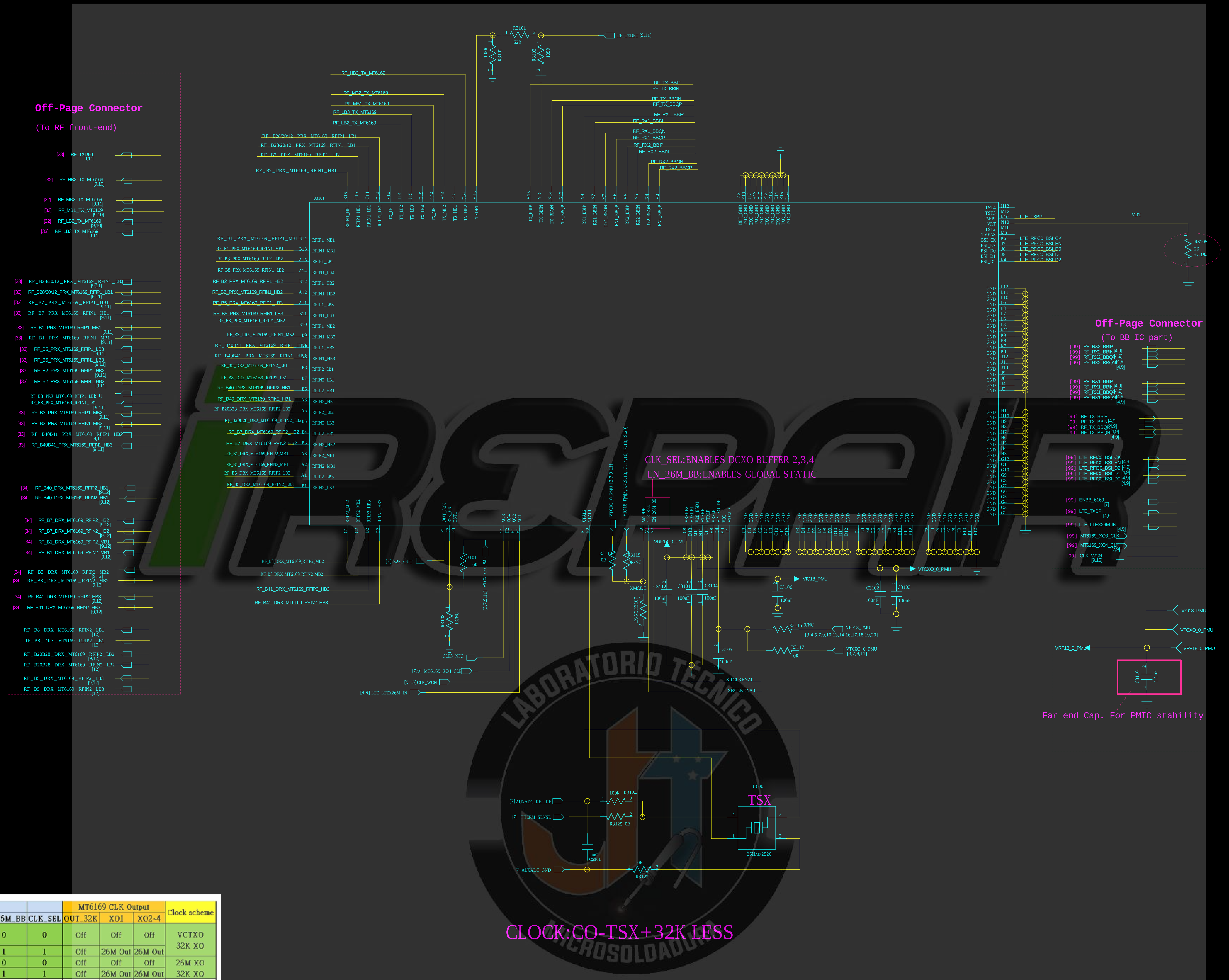
<Code>	A1	<Drawing Number>	1_13

SCALE: <Scale> SHEET: 8 OF 21



Case	MT6169 Control Pin					MT6169 CLK Output			Clock scheme
	XMODE	32K_EN	VDXCO_DIG	EN_26M_BB	CLK_SEL	OUT_32K	XO1	XO2~4	
1	0	0	VIO18	0	0	Off	Off	Off	VCTXO 32K XO
2	0	0		1	1	Off	26M Out	26M Out	
3	VIO18	0		0	0	Off	Off	Off	26M XO
4		0		1	1	Off	26M Out	26M Out	32K XO
5	VTCXO28	VTCXO28	VTCXO28	0	0	On	Off(LPM)	Off	26M XO
6				1	1	On	26M Out	26M Out	32K Less

Clock Scheme	32K Crystal Component				32K_EN			XMODE			VDCXO_DIG		VCTCXO / TSX Component										MT6328 AuxADC			VTCXO2 Power
	R2027	C2033	C2034	X2001	R3116	L3101	R3108	R3119	R3118	R3107	R3115	R3117	R3114	R3126	R3124	R3125	R3127	X3102	U3103	C3113	C3114	C3115	C3161	C2024	C2023	R2007	VTCXO2 Power Connection
VCTXO + 32K Exist	NC	22pF	22pF	32K X'tal	NC	NC	1K	NC	NC	1K	0R	NC	0R	0R	NC	NC	NC	NC	VCTCXO	100nF	100nF	NC	NC	0R	1uF	0R	Connect to VTCXO_0_PMU
TSX + 32K Exist	NC	22pF	22pF	32K X'tal	NC	NC	1K	0R	NC	NC	0R	NC	NC	NC	100K	0R	0R	TSX	NC	NC	NC	NC	NC	1nF	1uF	Bead	Connect to VTCXO_0_PMU
TSX + 32K Less	0R	NC	0R	NC	NC	0R	NC	NC	0R	NC	NC	0R	NC	NC	100K	0R	0R	TSX	NC	NC	NC	NC	NC	1nF	1uF	Bead	Connect to VTCXO_1_PMU

COMPANY: Motorola
TITLE: XT1758
DRAWN: <Drawn By> DATED: <Drawn Date>
CHECKED: <Checked By> DATED: <Checked Date>
QUALITY CONTROL: <QC By> DATED: <QC Date>
RELEASED: <Released By> DATED: <Release Date>
CODE: <Code> SIZE: A1 DRAWING NO: <Drawing Number> REV: 1_13
SCALE: <Scale> SHEET: 9 of 21

REVISION RECORD
LTR: ECO NO: APPROVED: DATE:
Off-Page Connector
(To RF front-end)
RF_HB2_TX_MT6169
RF_MB1_TX_MT6169
RF_LB2_TX_MT6169
RF_B5_PA_DPX
RF_B3_PA_DPX
RF_B2_PA_DPX
RF_B1_PA_DPX
RF_B7B41_PA_DPX
RF_B17_PA_DPX
RF_B17B20_PA_DPX
RF_B8_PA_DPX
AUX_IN1_NTC
MIPI0_SDATA
MIPI0_SCLK
VPA_PMU
LTE_VMIPI
VIO18_PMU
RF_B20/B28A_PA_DPX
RF_B28B_PA_DPX
RF_B12_PA_DPX
RF_B4(LANA)_PA_DPX
RF_B40_TRX_TXM
RF_B38B41_TRX_TXM
RF_B20/B28A_PA_DPX
RF_B28B_PA_DPX
RF_B5_PA_DPX
RF_B8_PA_DPX
RF_B4(A)_PA_DPX
RF_B3_PA_DPX
RF_B2_PA_DPX
RF_B1_PA_DPX
VPA_PMU
VPA_VCC
PIN20=LB5
RFMD and Murata have pin to pin products. Please check latest QVL and matching accordingly
4G PA input need LPF for harmonic rejection
3/4G_PAIN_LB
3/4G_PAIN_MB
3/4G_PAIN_HB
VBAT
VC7643
LPF for more margin on Tx spurious
B40 TRX
B7B41 TX
B38B41 TRX
Thermistor / To sense board level temperature
390K 1%
NTC3201 close to PA , and located in the same layer
Power Net Connection
COMPANY: Motorola
TITLE: XT1758
DRAWN: <Drawn By> DATED: <Drawn Date>
CHECKED: <Checked By> DATED: <Checked Date>
QUALITY CONTROL: <QC By> DATED: <QC Date>
RELEASED: <Released By> DATED: <Release Date>
CODE: <Code> SIZE: A1 DRAWING NO: <Drawing Number> REV: 1_13
SCALE: <Scale> SHEET: 10 21

TRX3,4,5,8,9不能走低频信号
SP14T
Main_ANT_704~2690MHz
ASM_Main
VC7916
MT6169 TX Output need a DC block(MUST).
2G_PAIN_HB
2G_PAIN_LB
Pi attenuation for LB 8PSK TX in RX band noise
B2 TRX
B5 TRX
B40/B41 PRX
此处有跳线
B8 TRX
B3 TRX
B7 TRX
B4 TRX for LA
BAND28A
BAND28B/Band20
B1 TRX
LANA B4双工器 SAYEY1G73CA0F0A
For power VTCXO2 star connection
Motorola
XT1758
<Code>
A1
<Drawing Number>
1_13


DRX ANT: 704~2690MHz
DRX Car Kit
SP8T
B7 DRX
B1&B4 DRX
B8 DRX
B38 DRX
B40 DRX
B3 DRX
B5 DRX
B28 B20 DRX
B40 for india B2 for LA DRX
Motorola
XT1758
<Code>
A1
<Drawing Number>
1_13
<Drawn By>
<Drawn Date>
<Checked By>
<Checked Date>
<QC By>
<QC Date>
<Released By>
<Release Date>
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21

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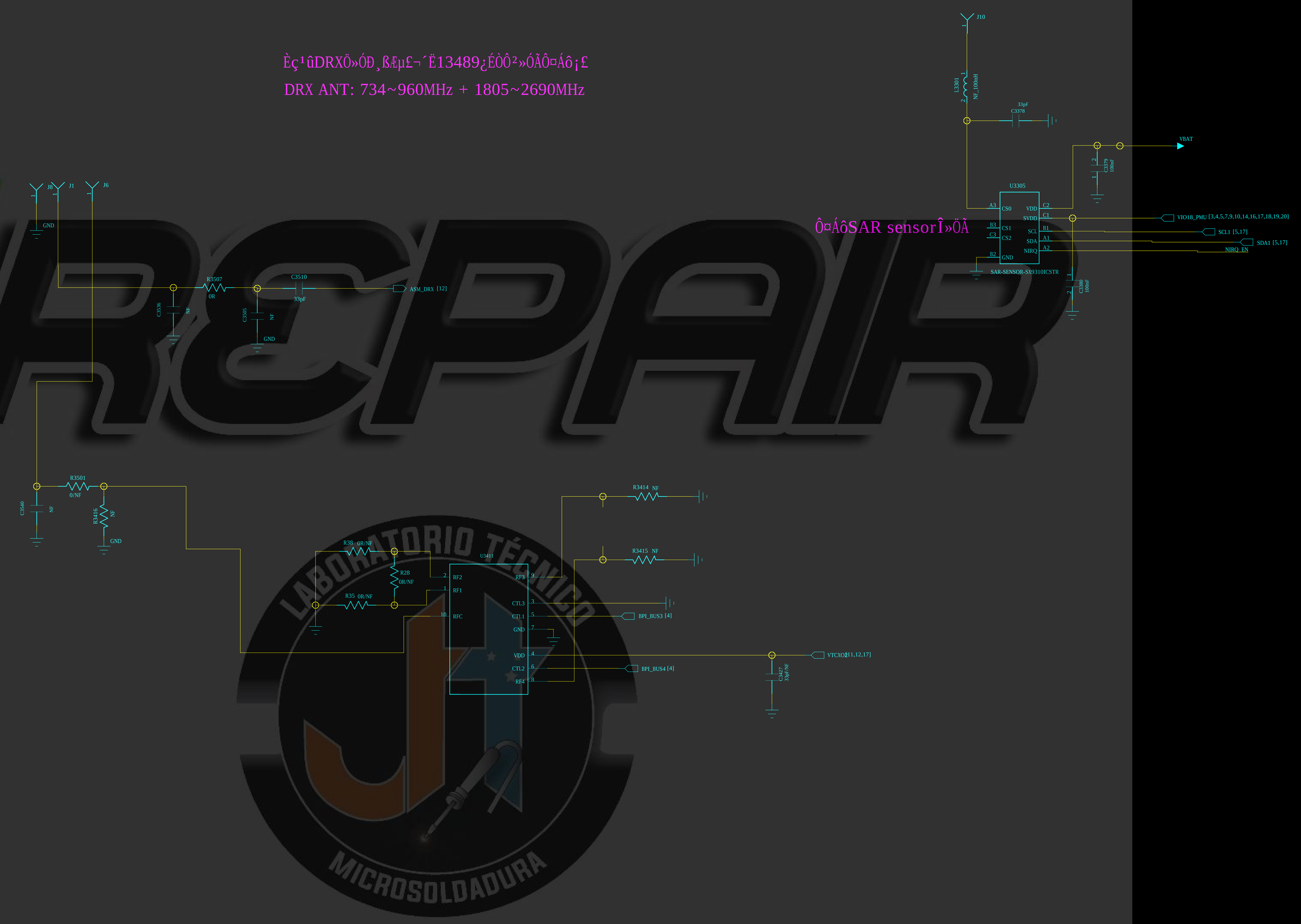
DRX ANT: 734~960MHz + 1805~2690MHz

ÔÝÁôSAR sensorÎ»ÖÃ



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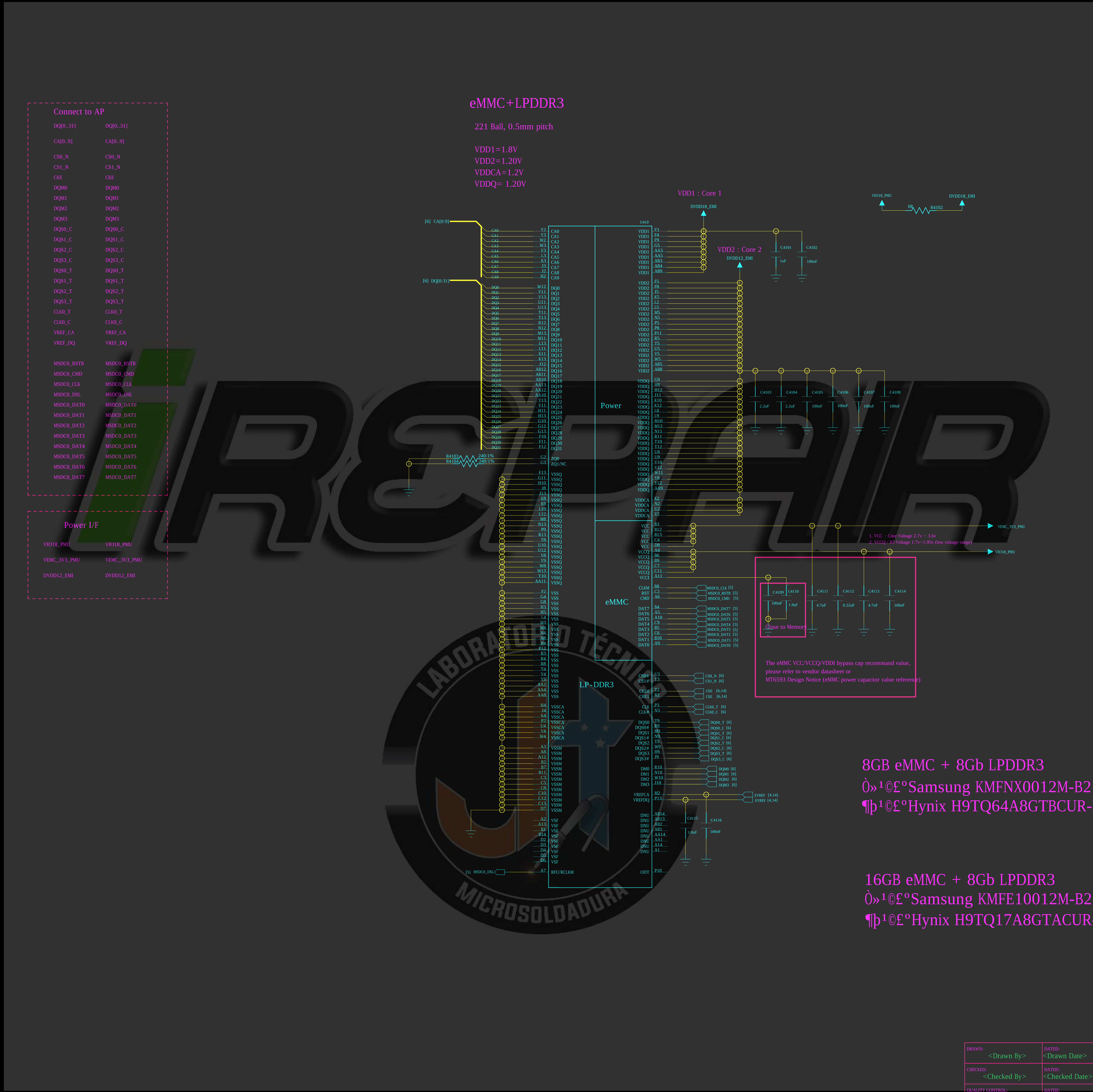
DATED:	<Release Date>



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CHECKED:	<Checked By>
DATED:	<Checked Date>
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DATED:	<QC Date>
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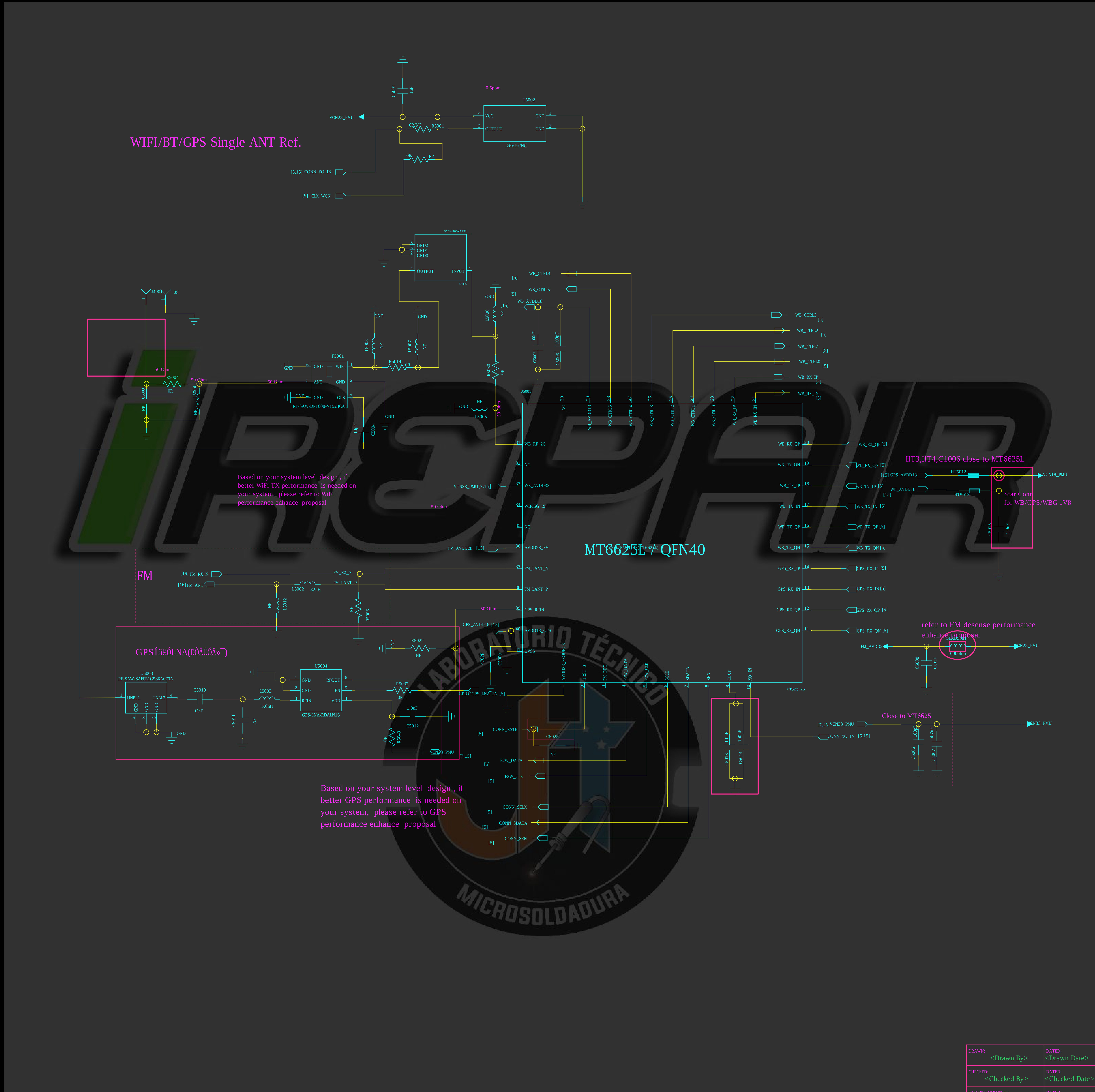
DATED:	<Release Date>



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QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

COMPANY: Motorola

TITLE: XT1758

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A1	<Drawing Number>	1_13

SCALE: <Scale> SHEET: 15 21

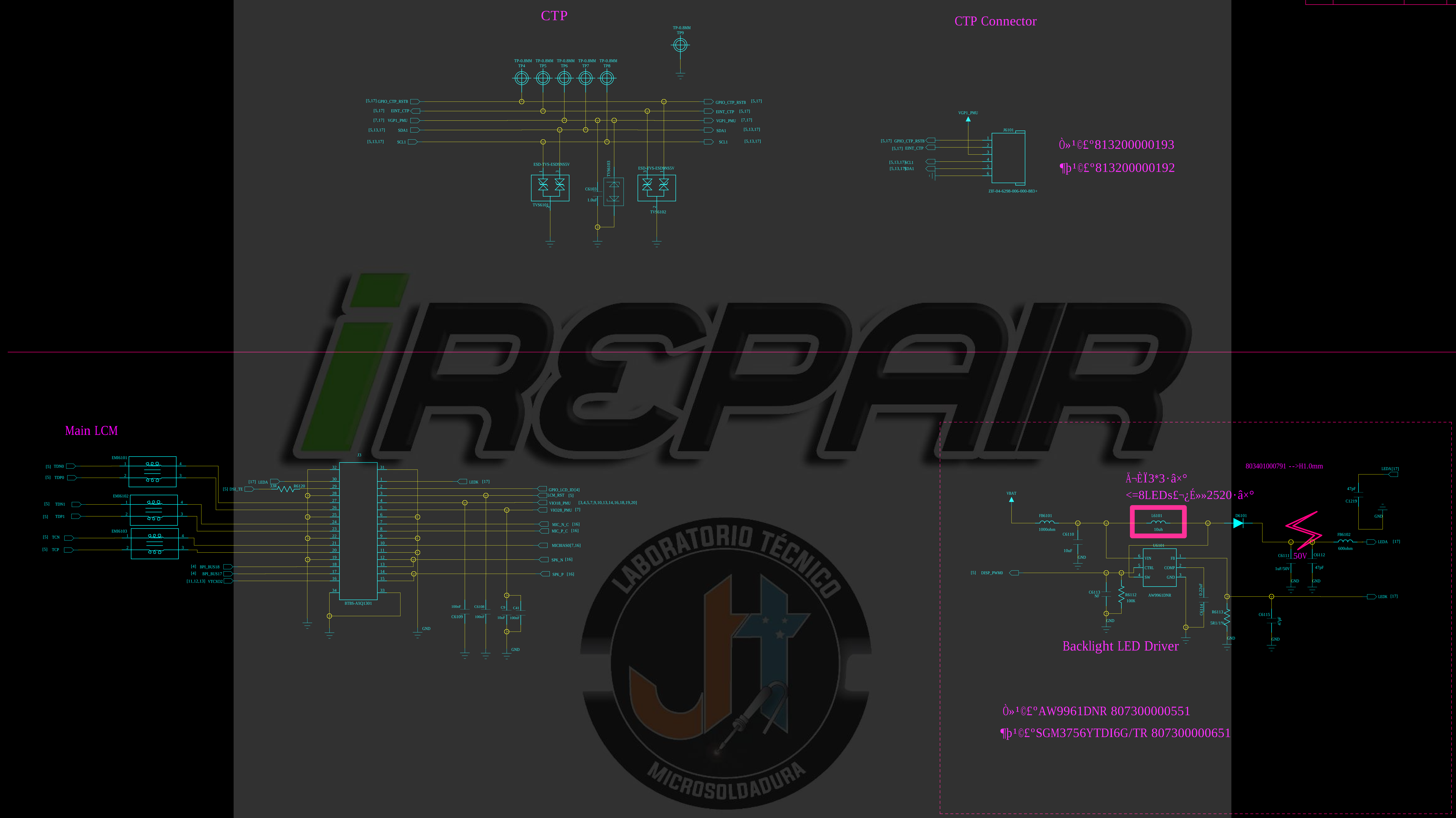
Schematic design notice of "60_PERI_AUDIO_IO" page.

Note 60-1: The equivalent capacitance of audio and speech ESD protection device must be <=330pF. choose bi-directional device only

Note 60-2: The equivalent capacitance of FM ANT. ESD protection device must be <=1pF.

Note 60-3: TVS Stand off voltage for speaker should >=5V

COMPANY:	Motorola
TITLE:	XT1758
CODE: <Code>	SIZE: A1
DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>	SHEET: 16 of 21



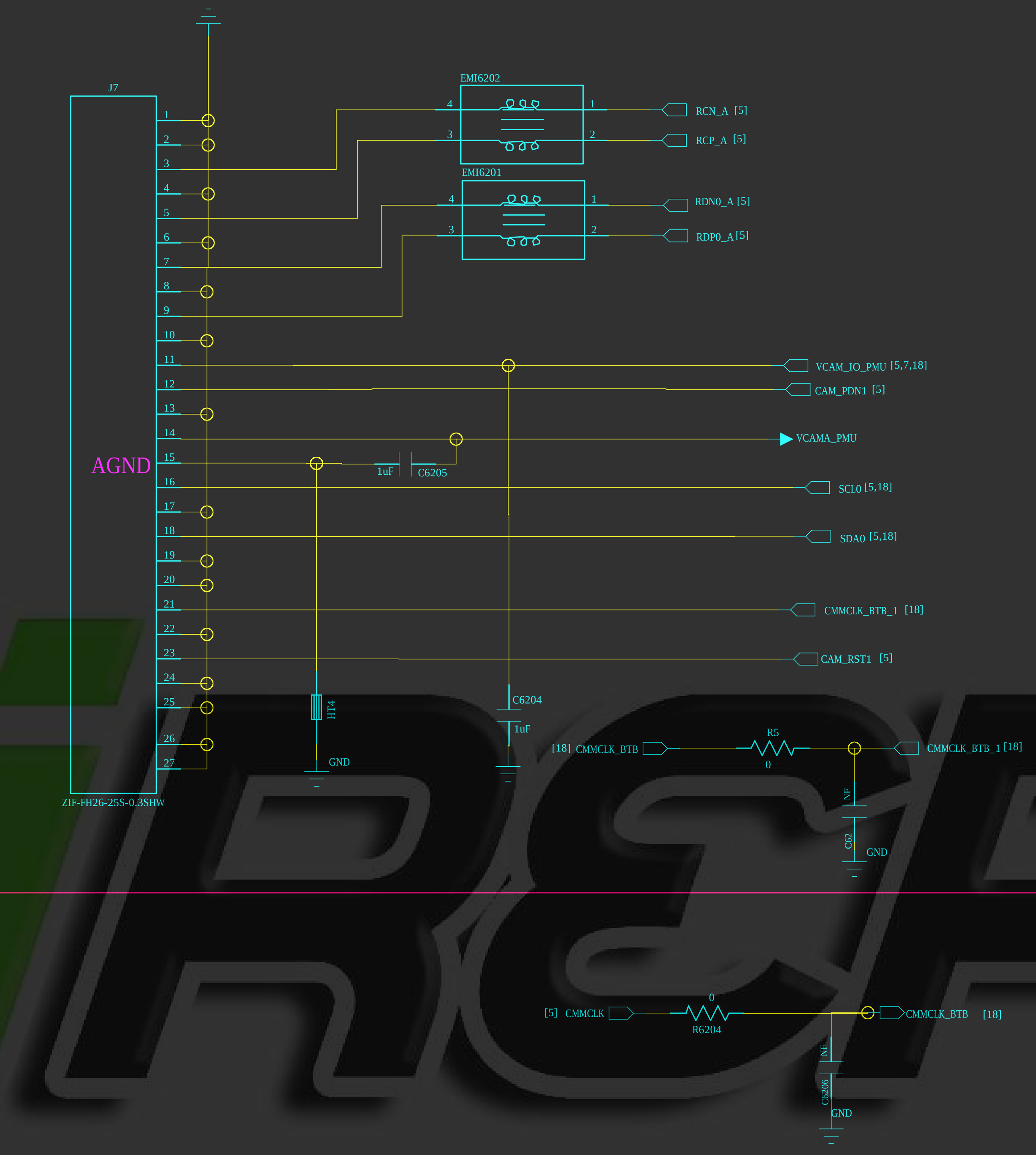
Schematic design notice of "61_PERI_LCD_CTP" page.

Note 61-1: GT1151 I2C address: 0X5D (Write:0xBA, Read:0xBB) or 0x14 (Write:0x28, Read:0x29)

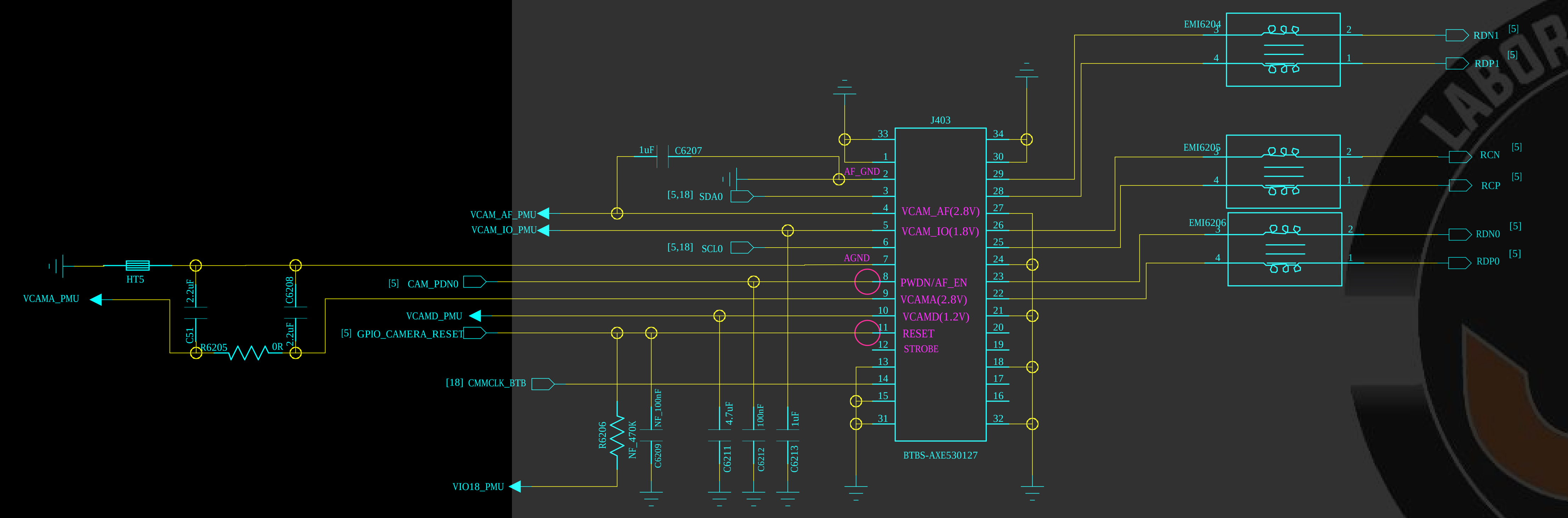
DRAWN: <Drawn By>	DATED: <Drawn Date>	COMPANY:	Motorola
CHECKED: <Checked By>	DATED: <Checked Date>	TITLE:	XT1758
QUALITY CONTROL: <QC By>	DATED: <QC Date>	CODE: <Code> SIZE: A1	DRAWING NO: <Drawing Number> REV: 1_13
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>	SHEET: 17 of 21

Front Camera 2M_FF

DOVDD=1.8V
AVDD=2.8V



Main Camera 5M_AF



FLASH LED

Front

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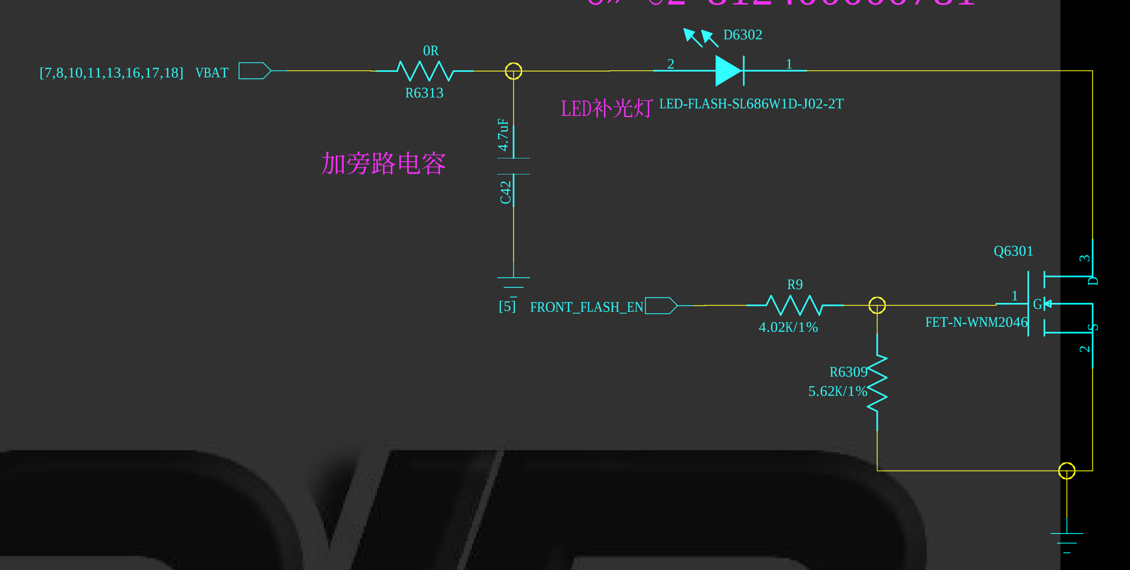
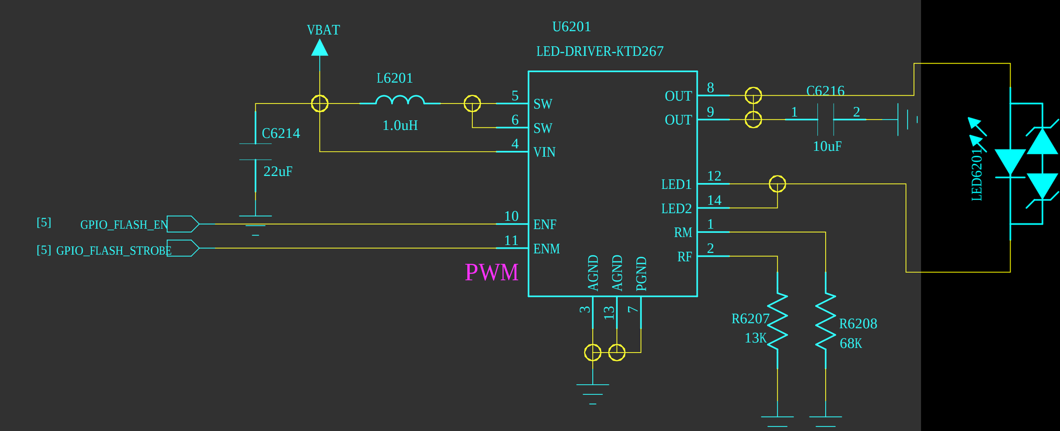


Back



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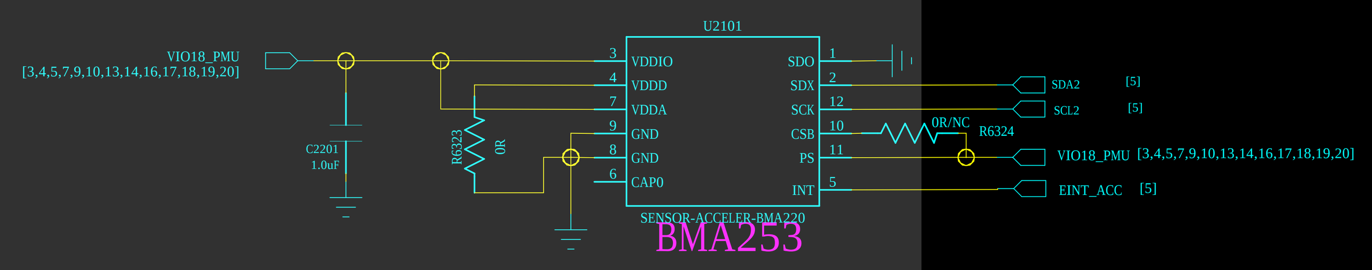
Ò»¹©£°OCP8132AVAD 807300000581
¶þ¹©£°SYWT78DUC 807300000461

flashing:Iled1=Iled2=6800/9.1k
torch:Iled1=Iled2=6800/68k

DRAWN: <Drawn By>	DATED: <Drawn Date>	COMPANY:	Motorola
CHECKED: <Checked By>	DATED: <Checked Date>	TITLE:	XT1758
QUALITY CONTROL: <QC By>	DATED: <QC Date>	CODE: <Code> SIZE: A1 DRAWING NO: <Drawing Number>	REV: 1_13
RELEASED: <Released By>	DATED: <Release Date>	SCALE: <Scale>	SHEET: 18 21

M-Sensor

G-Sensor



BMA: R6323 0 R6324 NF
BMC156: R6323 0 R6324 0
KXTJ2-1009: R6323 NF R6324 0

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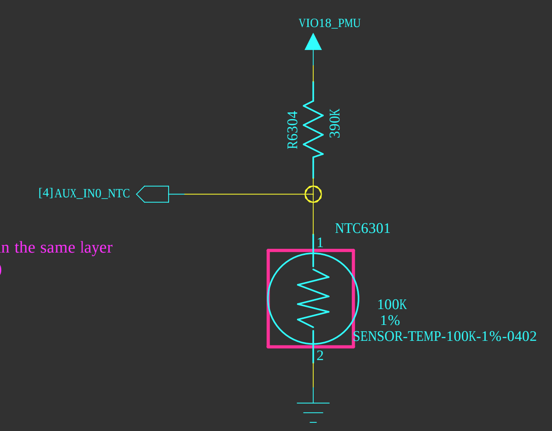
Accerometer

ALS+PS+IR

Thermistor



NTC6301 close to BB ,and located in the same layer
(can refer to thermal design notice)

Thermistor / To sense board level temperature

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

COMPANY: Motorola			
TITLE: XT1758			
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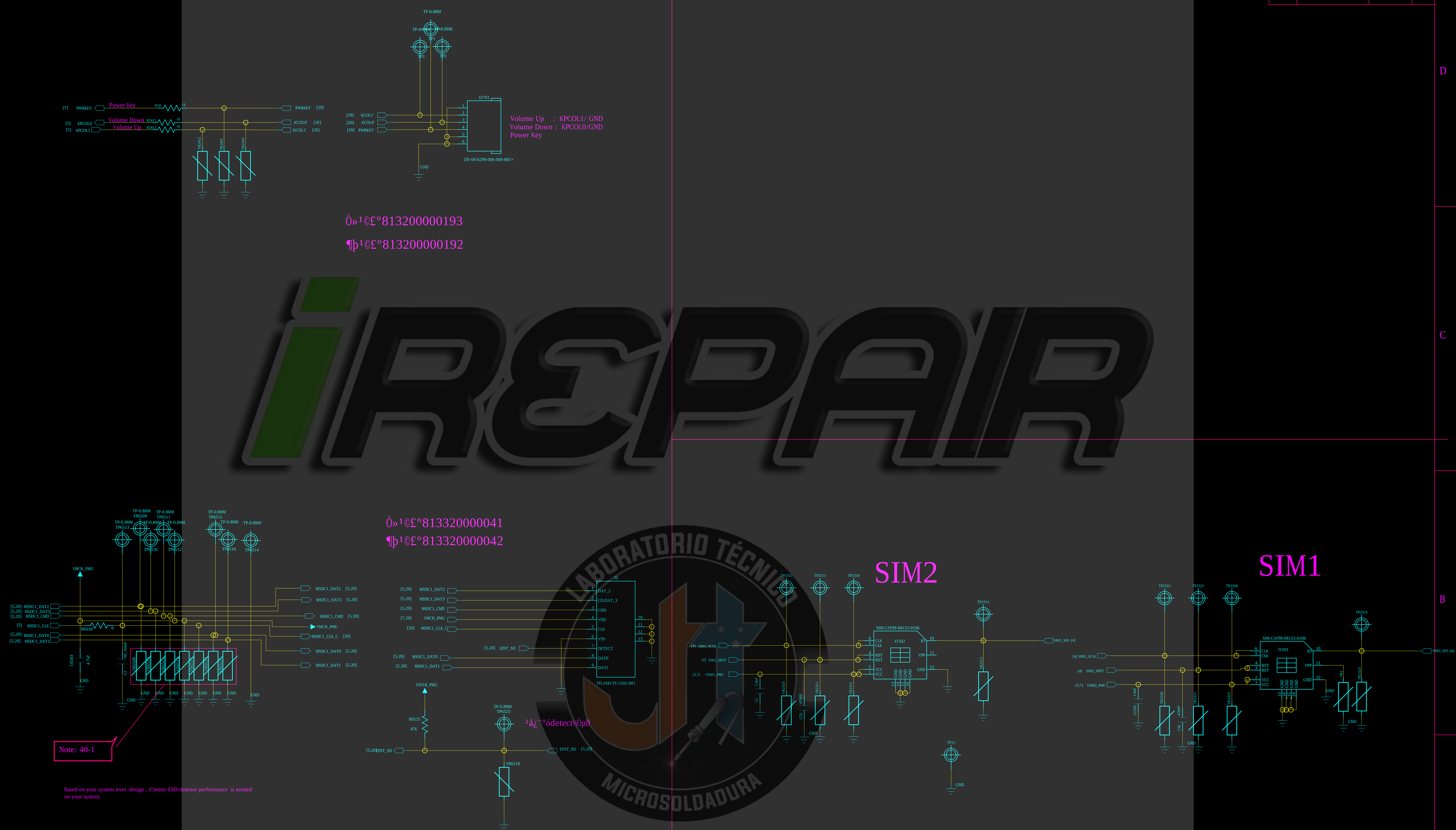

Schematic design notice of "40_MEMORY_SD Card" page.

Note 40-1: The equivalent capacitance of MSDC ESD protection device must be <=10pF.
But for NFC app. equivalent capacitance of MSDC_NFC_SWPIO and MSDC_NFC_VCCSWP should <=0.5pF.

COMPANY:	Motorola
TITLE:	XT1758
CODE:	<Code>
SIZE:	A1
DRAWING NO:	<Drawing Number>
REV:	1_13
SCALE:	<Scale>
SHEET:	20 21
DRAWN:	<Drawn By>
DATED:	<Drawn Date>
CHECKED:	<Checked By>
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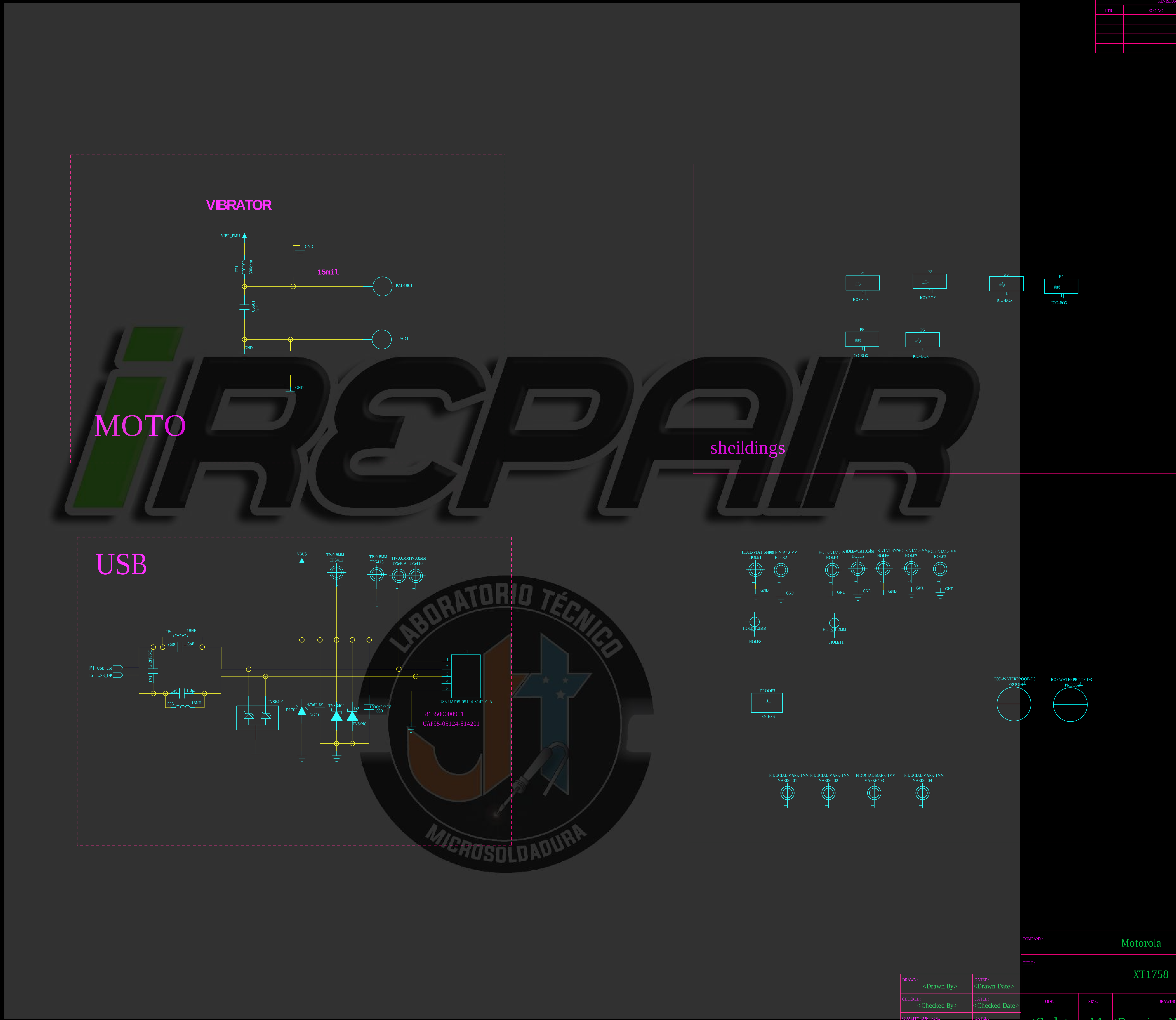

COMPANY:	Motorola
TITLE:	XT1758
CODE:	<Code>
SIZE:	A1
DRAWING NO:	<Drawing Number>
REV:	1_13
SCALE:	<Scale>
SHEET:	21

DRAWN:	<Drawn By>	DATED:	<Drawn Date>
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QUALITY CONTROL:	<QC By>	DATED:	<QC Date>
RELEASED:	<Released By>	DATED:	<Release Date>